ITA03-MP Assignment 2

MOBILE COMPUTING

ASSIGNMENT 2 – CO2

CO2 Coverage: Mobile Telecommunication Systems; Multiple Access Techniques (SDMA, TDMA, FDMA, CDMA, OFDMA); GSM Architecture, Services, Radio Interfaces, Localization, Calling, Handover & Security; Advanced GSM Extensions (GPRS, HSCSD, DECT).

Nature of Assignment: Individual Duration: 1 Week
Total Marks: 50
Submission Date: 06/11/26

Q1 – Mobile Communication or Mobile Computing? 5 Marks

Problem Context – Field-Service Technology Audit

A utility company equips its field staff with: (a) a ruggedised tablet fetching work orders over cellular data, (b) a laptop plugged into Ethernet at different offices, (c) a fixed wireless router at a remote substation, (d) a push-to-talk handset, and (e) a tablet app that records meter readings offline and synchronises later.

Your Challenge

Investigate whether “wireless” and “mobile” mean the same thing, and whether mobile communication and mobile computing are the same thing.

Before analysis:

- Define at least four of your own classification criteria (e.g., user mobility, wireless link, tolerance of disconnection, location/context awareness, device resource limits).
- Predict how each of the five cases is classified (fixed-wired, nomadic, fixed-wireless, mobile) and whether it is mainly a mobile-communication or a mobile-computing problem, giving reasons.

- Design one additional ambiguous case of your own to test your criteria.

Test Your Reasoning

Build a small simulation of case (e) with link availability of 90%, 60% and 30%. Compare two strategies: send immediately (fail if disconnected) vs store-and-forward queue. Record updates delivered, delayed and lost, and mean delivery delay.

Inquiry

Which characteristics of mobile computing are exposed by your results that a wireless link alone does not solve? Which device categories need application-level changes?

Evidence Expected: Own classification criteria, reasoned predictions, an engineered ambiguous case, simulation counts for both strategies, and a justified conclusion.

Q2 – Sharing the Channel – FDMA vs TDMA Under Different Traffic 6 Marks

Problem Context – Hostel-Campus Voice and Telemetry

A cell may use a 2 MHz band per direction. Option 1 is FDMA with 25 kHz channels. Option 2 is TDMA with 200 kHz carriers, each carrying 8 time slots. Both offer 80 nominal channels.

Your Challenge

Investigate whether equal nominal capacity means equal usable capacity.

- Estimate usable channels after guard bands, guard times, synchronisation and control overheads (state and justify your values).
- Predict which scheme gives better utilisation for (i) continuous voice, (ii) bursty data with about 10% duty cycle, and (iii) a 70/30 voice–data mix, and analyse why.

Design three contrasting traffic datasets, including one edge case of your own (e.g., very short calls or synchronised arrivals). Simulate dedicated-channel allocation under both schemes and record utilisation, blocking and wasted capacity.

Inquiry

Where do the two schemes diverge, and what causes it? Design

Decision

At a campus telemetry site, 90% of devices send a 200-byte report per minute. Is dedicated FDMA/TDMA channel allocation justified? Defend your answer.

Evidence Expected: Capacity calculations with overheads, reasoned predictions, three designed traffic conditions, utilisation/blocking evidence and a justified decision.

Q3 – CDMA – Orthogonality, Spreading and the Near-Far Problem 7 Marks

Problem Context – Crowded Stadium Uplink

Several handsets transmit simultaneously in the same band to one base station, each spreading its data with its own code.

Your Challenge

Design a direct-sequence CDMA uplink simulation. Determine how data bits are spread, how the receiver despreads and decides, and what the processing gain is.

Before executing, predict the bit-error behaviour as the number of users grows for: (i) orthogonal (e.g., Walsh) codes with synchronous users, (ii) random non-orthogonal codes of the same length, and (iii) orthogonal codes where one user transmits at ten times the power. analyse your reasoning.

Implement and test with at least three code lengths (e.g., 8, 16, 64). Record bit errors over a stated number of bits, correlation values and processing gain.

Inquiry

- Why do orthogonal codes still fail when users are unsynchronised or powers are unequal?
- Why is power control essential in CDMA?
- Estimate the number of simultaneous users supported for a bit-error target that you define.

Evidence Expected: A working DS-CDMA simulation, predictions made before execution, bit-error evidence for three code lengths and three conditions, and a reasoned explanation of the results.

Q4 – OFDMA – Giving Each User the Subcarriers They Can Use Best 6 Marks

Problem Context – Lecture-Hall Downlink

Eight users share 64 subcarriers. Each user's channel quality varies across subcarriers (frequency-selective fading).

Your Challenge

- Before designing, determine how subcarrier spacing relates to symbol duration, why a cyclic prefix is needed, and verify orthogonality of at least three subcarriers numerically (FFT or inner product).
- Design three allocation approaches: fixed contiguous blocks, interleaved round-robin, and channel-aware best-subcarrier.
- Predict total throughput and fairness (e.g., Jain's index) of each, with reasons.

Implement using generated fading gains with fixed seeds over at least 1,000 channel realisations. Record sum throughput, per-user throughput and fairness.

Inquiry

Why can maximising total throughput starve some users? How does the number of users change the benefit of channel-aware allocation?

Design Decision

Select an allocation for a lecture hall in which every student must receive at least a minimum rate that you define and justify.

Evidence Expected: Numerical orthogonality verification, predictions, three allocation schemes implemented, throughput and fairness results, and a justified allocation choice.

Q5 – SDMA – Reusing the Same Channel in Different Directions 6 Marks

Problem Context – Highway Cell With Directional Antennas

Vehicles along a highway use the same frequency and time slot. The base station can serve them with an omni antenna, a 120° sector antenna, or an N-element uniform linear array steered toward each user.

Your Challenge

- Derive the array factor of an N-element uniform linear array and predict how main-lobe width and side-lobe level change with N.
- Define an SIR threshold and predict the minimum angular separation at which two users can share the same resource.
- Compute patterns for N = 2, 4, 8, 16 and evaluate the SIR at user B when the beam is steered at user A, for separations of 5°–60°. Record beamwidth, side-lobe level and the minimum separation meeting your threshold.

Inquiry

What limits SDMA in practice (angular spread, clustered users)? Why can two users in nearly the same direction not be separated?

Design Decision

At a toll plaza most vehicles arrive in the same direction. Would you rely on SDMA there? Defend your decision.

Evidence Expected: Array-pattern derivation, predictions, computed patterns and SIR results across angular separations, and a reasoned explanation of SDMA limits.

Q6 – Choosing Multiple-Access Techniques for Real Deployments 7 Marks

Problem Context – Four Deployments

Choose access methods for: (1) wide-area cellular voice, (2) a jam-resistant tactical link, (3) a high-rate stadium downlink, (4) a network of 5,000 low-rate sensors sending small reports.

Your Challenge

- Define at least five weighted criteria and justify the weights yourself.
- Rate each of SDMA, TDMA, FDMA, CDMA and OFDMA for each deployment using your own Q2–Q5 evidence wherever possible.
- Choose a primary technique and, where useful, a hybrid for each deployment.

- Perform a sensitivity check by changing weights by ±20%: does any choice change?

Inquiry

Why do real systems combine techniques (for example FDMA with TDMA)? Which of your choices is least robust, and why?

Evidence Expected: Independently justified criteria and weights, ratings traceable to your own Q2–Q5 evidence, sensitivity analysis and defended choices.

DAY-4 DESIGN CHALLENGE

Q7 – Diagnose, Redesign and Validate 13 Marks

Review the evidence you generated while working on Q1–Q6.

Select ONE solution that demonstrates a significant limitation as its problem size or operating requirement increases. Examples of limitations you may find in this variant:

- Dedicated FDMA/TDMA channels sit idle under bursty traffic.
- A CDMA link collapses under power imbalance or loss of synchronisation.
- Best-subcarrier OFDMA allocation starves weak users.
- SDMA fails when users are angularly clustered.

You are not permitted simply to replace the solution with a more advanced technology or standard and state that it is better. Your task is to understand the problem first and then develop an appropriate redesign.

A. Diagnose the Bottleneck 2 Marks

- Where most capacity, signalling or computational effort is being spent.
    - Which counted quantity grows significantly as load or size increases.
- Whether resources are idle, wasted or repeated unnecessarily.
- Which characteristic of the original design causes the limitation.

Evidence Expected: A precise diagnosis supported by evidence from your original implementation and analysis.

B. Define the Design Objective 2 Marks

- State what your redesign must achieve before proposing it (e.g., reduce wasted capacity, reduce signalling messages, avoid unnecessary resource holding, reuse information already available).
- Specify a measurable objective wherever possible.

Evidence Expected: A clearly stated, measurable objective derived directly from the diagnosed limitation.

C. Design the Revised Strategy 4 Marks

- What you have changed and what property or information the new design exploits.
- Which operations or resource usages are avoided or reduced.
- Whether additional state, preprocessing or complexity is introduced, and any new trade-offs.
- Provide pseudocode, a flowchart or a message-sequence design.
- Marks are based on the quality of reasoning, not merely on naming a known technique.

Evidence Expected: A coherent redesign directly addressing the bottleneck, with a design representation and technical justification.

D. Predict Before Implementing 1 Mark

- Predict how the redesigned solution should differ from the original and in which measure you expect improvement.

Evidence Expected: A clear, testable prediction made before experimental validation.

E. Implement and Validate the Redesign 2 Marks

- Implement a prototype or simulation and compare Original → Redesigned under the same or equivalent test conditions.
- Use counted measures (e.g., slots used, messages, blocking, bit errors, throughput, fairness); run-time is supporting evidence only.

Evidence Expected: Comparable experimental evidence for the original and redesigned solutions.

F. Defend the Final Design 2 Marks

- Was the prediction supported, and what was the magnitude of improvement?
- What new costs or trade-offs appeared, and when might the original still be preferable?
- A redesign that does not outperform the original is not automatically unsuccessful if you correctly analyse why the hypothesis failed.

Evidence Expected: A critical, evidence-based evaluation and a defensible final engineering conclusion.

Mark Distribution

Question	Assessed Area	Marks
Q1	Mobile communication vs. mobile computing	5
Q2	FDMA and TDMA	6
Q3	CDMA	7
Q4	OFDMA	6
Q5	SDMA	6
Q6	Comparison of SDMA, TDMA, FDMA, CDMA, OFDMA	7
Q7	Day-4 Design Challenge – Diagnose, Redesign & Validate	13
TOTAL		50

Submission Requirements (Digital)

- Assignment report (PDF) containing the required evidence for Q1–Q7. No handwritten work.
- Source code for all implementations/simulations (GitHub repository link with README and fixed random seeds).

- Raw experimental data (CSV) used to produce your results.
- Tables and graphs used in your analysis.
- Counted quantities (slots, messages, bits, blocked calls, codes, throughput) are primary experimental evidence. Execution time may be reported as supporting evidence but must not be the sole basis for conclusions.
- Standards parameters (frame durations, rates, header sizes, etc.) must be obtained from cited references; state assumptions wherever a value is assumed.

Assessment Verification

You may be asked to analyse any part of your design, code, test-data design, experimental results, predictions, analysis and Design Challenge in a brief verification interpretion/viva. Disclose any AI or reference tools used. You must be able to analyse and defend the evidence as your own work.

Overall Assessment Expectation

This assignment does not primarily assess your ability to reproduce textbook descriptions of mobile systems. It assesses your ability to:

Predict → Design → Implement → Measure → Analyse → Evaluate → Diagnose → Redesign → Validate

Your conclusions and design decisions must be supported by your own reasoning and experimental evidence.

MOBILE COMMUNICATION AND MULTIPLE ACCESS TECHNIQUES

ENGINEERING INVESTIGATION, SIMULATION AND DESIGN REPORT

Submitted by: K. Goutham

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Department: B.E-CSE

Course: Mobile Computing

Course Code: ITA 0327

This report combines conceptual analysis, mathematical calculations, reproducible Python simulations, measured results and engineering decisions for Questions 1–7.

Abstract

This report investigates the distinction between mobile communication and mobile computing and evaluates five multiple-access techniques: FDMA, TDMA, CDMA, OFDMA and SDMA. Each question combines definitions and design reasoning with calculations or a reproducible simulation. The final design challenge diagnoses inefficient dedicated resource reservation for low-duty-cycle telemetry and evaluates an event-driven shared-resource redesign. Assumptions are stated explicitly so that results can be reproduced and interpreted within their limits.

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- Q1 – Mobile Communication or Mobile Computing?
- Q2 – Sharing the Channel: FDMA vs TDMA
- Q3 – CDMA: Orthogonality, Spreading and Near-Far
- Q4 – OFDMA: Subcarrier Allocation
- Q5 – SDMA: Directional Antennas and Spatial Reuse
- Q6 – Choosing Multiple-Access Techniques
- Q7 – Diagnose, Redesign and Validate
- Overall Conclusion
- Appendix A – Reproducible Python Code
- References

General methodology and reproducibility

Python 3 with NumPy and Matplotlib is used for numerical experiments. Random experiments use fixed seeds. Tables in this report are generated from the stated models. A simulation result is evidence for its model and assumptions; it is not a claim that every real network behaves identically.

Important modelling boundaries: (i) Q2 uses a simplified dedicated-resource event model and separately explains why a genuinely dynamic TDMA scheduler can exploit burstiness; (ii) Q3's synchronous matched-filter model isolates code correlation and near-far effects, and does not model a complete receiver; (iii) Q4 uses an idealised frequency-selective Rayleigh channel and equal per-subcarrier power; (iv) Q5 uses an ideal half-wavelength ULA in free space. These limitations are discussed rather than hidden.

Q1 – Mobile Communication or Mobile Computing?

1.1 Concepts and classification criteria

Mobile communication is the wireless exchange of information between devices or network nodes. Mobile computing is the use of portable computing devices and applications while the user or device changes location, often under intermittent connectivity and limited battery, storage or processing resources. Wireless describes the transmission medium; mobile describes mobility. The terms overlap but are not interchangeable.

Criterion	Meaning
User/device mobility	Can the user/device move during operation, or is it installed at one location?
Access medium	Is the network access wired, wireless, or mixed?
Connection continuity	Must the network remain continuously available?
Offline capability	Can useful work continue and synchronise later?
Context/resource constraints	Does location, battery, storage or limited processing affect operation?

1.2 Classification of the five cases

Case	Classification	Main emphasis	Reason
(a) Rugged tablet over cellular	Mobile	Both	Portable device moves; cellular link provides mobile communication and the tablet processes work orders.
(b) Laptop using Ethernet at offices	Nomadic	Mobile computing	Portable between locations but uses a fixed wired connection at each office; not continuously connected while moving.
(c) Fixed wireless router	Fixed-wireless	Communication	Wireless access is used, but the router and endpoint are fixed at the substation.
(d) Push-to-talk handset	Mobile	Mobile communication	Portable handset supports voice/group communication while staff move.
(e) Offline meter-reading tablet	Mobile	Mobile computing	Records and stores readings locally, then synchronises when connectivity returns.

1.3 Additional ambiguous case

Case (f): A field engineer carries a laptop and connects it to a smartphone hotspot. The laptop is mobile/nomadic depending on whether it is used while moving or only after stopping. The access link is wireless, but the computing application may continue offline. This case shows why mobility, wireless access and offline capability must be classified separately.

1.4 Simulation model and predictions

The experiment generates 1,000 updates, one per minute, with independent link availability at 90%, 60% and 30%. Strategy A sends immediately and loses an update if the link is unavailable at its generation time. Strategy B stores updates locally and, whenever a link becomes available, batch-synchronises all pending updates. Batch transfer is assumed to have sufficient capacity to clear the queue in one available interval. Updates still queued at the end are pending, not lost. This explicit batch assumption is important: a one-update-per-available-minute link would produce a different queueing result.

Prediction: immediate delivery success will fall with availability. Store-and-forward should deliver more updates, but mean delay and pending backlog should increase as availability falls.

1.5 Python implementation

```
import random
import statistics

def simulate(availability, total=1000, seed=42):
    rng = random.Random(seed)
```

```
    link = [rng.random() < availability for _ in range(total)]

    # Immediate send: failure at generation time means loss
    immediate_delivered = sum(link)
    immediate_lost = total - immediate_delivered

    # Store-and-forward: batch all pending updates at next online minute
    pending = []
    delays = []
    for t, online in enumerate(link):
        pending.append(t)
        if online:
            delays.extend(t - generated for generated in pending)
            pending.clear()

    return {
        "immediate_delivered": immediate_delivered,
        "immediate_lost": immediate_lost,
        "queued_delivered": len(delays),
        "queued_delayed": sum(d > 0 for d in delays),
        "pending_at_end": len(pending),
        "mean_delay_min": statistics.mean(delays) if delays else 0
    }

for p in (0.90, 0.60, 0.30):
    print(p, simulate(p))
```

1.6 Simulation results

Availability	Immediate delivered	Immediate lost	Queued delivered	Queued delayed	Pending at end	Mean delay (min)
90	892	108	1000	108	0	0.115
60	580	420	998	418	2	0.72
30	295	705	996	701	4	2.387

Interpretation: queued delivery is greater because updates are retained until the next online interval. The mean delay increases as availability decreases. The pending column is not counted as lost; it represents updates that have not yet been delivered by the end of the finite observation window.

1.7 Inquiry and conclusion

The simulation exposes intermittent connectivity, offline operation, local persistence, delayed synchronisation and the need to manage pending data. A wireless link alone does not provide these application properties. The offline meter-reading tablet needs application-level storage, retry, duplicate protection and conflict handling. The cellular tablet may also need session recovery; a push-to-talk handset needs link recovery and call continuity; a fixed router mainly needs robust network connectivity. The experiment supports store-and-forward for delay-tolerant meter readings, provided the application handles duplicate uploads and eventual consistency.

Q2 – Sharing the Channel: FDMA vs TDMA Under Different Traffic

2.1 Nominal capacity and overhead assumptions

The stated 2 MHz band per direction gives 2,000 kHz. FDMA with 25 kHz channels provides 2,000/25 = 80 nominal channels. TDMA with 200 kHz carriers and eight slots provides (2,000/200) × 8 = 80 nominal time slots.

For a transparent planning estimate, assume FDMA reserves 10% of the band-equivalent capacity for guard bands, control and edge effects: 80 × 0.90 = 72 usable channel-equivalents. For TDMA, assume 10% of each frame's slot-time is consumed by guard time, synchronisation and control: 80 × 0.90 = 72 usable slot-equivalents. These are illustrative engineering assumptions, not universal standards values. Actual overhead depends on channel plan, modulation, frame format and control design.

Scheme	Nominal calculation	Assumed overhead	Estimated usable capacity
FDMA	2,000/25 = 80 channels	10% guard/control allowance	72 channel-equivalents
TDMA	(2,000/200)×8 = 80 slots	10% guard/sync/control allowance	72 slot-equivalents

2.2 Predictions for traffic types

- Continuous voice: FDMA offers a simple continuously available frequency channel. TDMA can also support voice efficiently when calls occupy scheduled slots, but requires timing synchronisation.
- Bursty data at 10% duty cycle: dynamic TDMA can assign slots only when data is ready, reducing idle reservation. Static FDMA dedicated to each user wastes capacity during silent periods.
- 70/30 voice-data mix: a hybrid or dynamic TDMA scheduler can reserve predictable service for voice while sharing remaining slots among data users. FDMA offers isolation but may strand unused capacity.

2.3 Designed traffic datasets and simulation

Three contrasting traffic conditions are modelled over a 60-minute interval: (1) continuous voice calls with mean holding time 3 minutes; (2) bursty reports with mean active time 0.25 minute; and (3) an edge case of synchronised short calls represented by a high arrival rate and 0.15-minute mean holding time. Arrivals and holding times are exponential. Each scheme is represented by 72 equivalent dedicated resources after overhead. This isolates blocking under equal resource counts; it does not simulate dynamic slot pooling.

Traffic dataset	Arrival rate (per min)	Mean holding (min)	What it tests
Continuous voice	0.85	3.0	Long-lived resource occupancy
Bursty data (10% duty approximation)	0.3	0.25	Short, intermittent resource use
Synchronised short-call edge case	2.0	0.15	Arrival bursts and transient blocking

2.4 Simulation implementation

```
import random

def simulate_resources(resources, arrival_rate, mean_hold,
                       horizon=60, seed=1):
    rng = random.Random(seed)
    t, arrivals = 0.0, []
    while True:
        t += rng.expovariate(arrival_rate)
        if t >= horizon:
            break
        arrivals.append((t, rng.expovariate(1.0 / mean_hold)))

    active_ends = []
    accepted = blocked = 0
    busy_resource_minutes = 0.0
    for arrival, hold in arrivals:
        active_ends = [end for end in active_ends if end > arrival]
        if len(active_ends) < resources:
            active_ends.append(arrival + hold)
```

```
            accepted += 1
            busy_resource_minutes += hold
        else:
            blocked += 1

    offered = len(arrivals)
    return {
        "offered": offered, "accepted": accepted, "blocked": blocked,
        "blocking_probability": blocked / offered if offered else 0,
        "utilisation": min(1, busy_resource_minutes /
                           (resources * horizon))
    }
```

2.5 Measured results

Traffic	Scheme	Offered	Accepted	Blocked	Blocking probability	Resource utilisation
Continuous voice	FDMA	43	43	0	0.000	2.73%
Continuous voice	TDMA	43	43	0	0.000	2.73%
Bursty data (10% duty)	FDMA	12	12	0	0.000	0.09%
Bursty data (10% duty)	TDMA	12	12	0	0.000	0.09%
Edge: synchronized short calls	FDMA	131	131	0	0.000	0.47%
Edge: synchronized short calls	TDMA	131	131	0	0.000	0.47%

Because the simplified dedicated model gives FDMA and TDMA the same 72-resource pool and the same arrival/holding assumptions, their numerical blocking and utilisation are equal. This is an intentional controlled comparison, not evidence that the real schemes are identical. The practical divergence occurs when TDMA dynamically reallocates idle slots among users, while a fixed FDMA assignment can leave a user's frequency idle. A full scheduler simulation would need per-user queues, slot request/grant signalling and frame timing.

2.6 Design decision and conclusion

For 5,000 telemetry devices sending one 200-byte report per minute, dedicated channel allocation is not justified. The reports are short and intermittent, so dedicating a continuously reserved channel to every sensor would create severe idle capacity. A shared random-access or scheduled aggregation design, with batching and collision/backoff control, is more appropriate. The exact access protocol depends on latency, reliability, battery and coverage requirements. FDMA remains useful for predictable continuous traffic and interference isolation; TDMA is attractive when slots can be assigned dynamically.

Q3 – CDMA: Orthogonality, Spreading and the Near-Far Problem

3.1 DS-CDMA principle

In direct-sequence CDMA, each information bit is multiplied by a user-specific spreading sequence of L chips. With BPSK, bit 1 is represented by +1 and bit 0 by −1. The receiver correlates the received chip vector with the intended user's code, integrates the result and decides by its sign. Processing gain is approximately Gp = L in linear units, or 10 log10(L) dB.

For code length L = 8, 16 and 64, the processing gains are 9.03 dB, 12.04 dB and 18.06 dB respectively.

3.2 Predictions before execution

- Synchronous orthogonal Walsh codes: ideal cross-correlation is zero, so multiuser interference is cancelled by a matched filter in a perfectly synchronised, noiseless channel. Errors should mainly be noise-limited.
- Random non-orthogonal codes: nonzero cross-correlations create multiple-access interference; error probability generally rises as the number of simultaneous users grows.
- Orthogonal codes with one user at ten times the power: in the perfectly synchronous ideal Walsh model, orthogonality still cancels the strong user's contribution. In practice, timing offsets, multipath and imperfect channel estimation destroy this ideal cancellation and cause near-far interference.

3.3 Simulation setup and results

The experiment uses BPSK, additive white Gaussian noise at 12 dB chip-level noise setting, 4,000 bits per user, and up to eight simultaneous users. Walsh codes are rows of a Hadamard matrix normalised to unit energy. Random codes are independent bipolar sequences. The near-far case gives one interfering user ten times the power (amplitude multiplied by √10). The model is synchronous and single-path; therefore the near-far Walsh result is an ideal lower-interference case.

Code length	Condition	Users	Bit errors	Total bits	BER	Max off-diagonal |correlation|
8	Synchronous Walsh	8	2	32000	0.000063	0.000
8	Random non-orthogonal	8	5141	32000	0.160656	0.750
8	Walsh near-far 10x power	8	0	32000	0.000000	0.000
16	Synchronous Walsh	8	2	32000	0.000063	0.000
16	Random non-orthogonal	8	4048	32000	0.126500	0.625
16	Walsh near-far 10x power	8	0	32000	0.000000	0.000
64	Synchronous Walsh	8	2	32000	0.000063	0.000
64	Random non-orthogonal	8	415	32000	0.012969	0.281
64	Walsh near-far 10x power	8	1	32000	0.000031	0.000

3.4 Implementation outline

```
import numpy as np

def hadamard(L):
    H = np.array([[1.0]])
    while len(H) < L:
        H = np.block([[H, H], [H, -H]])
    return H

# For each user: map bits to +/-1, multiply by its code.
# Add all users' chip vectors and AWGN.
# At the receiver: dot received chips with target code.
# Decide +1 for non-negative correlation, otherwise -1.
# BER = total bit errors / total transmitted bits.
```

The complete reproducible code is provided in Appendix A. It calculates the code correlation matrix and bit errors for each tested length and condition.

3.5 Inquiry and engineering interpretation

Orthogonality is a property of aligned code vectors over the same integration interval. If users are unsynchronised, the receiver's integration windows no longer cover the same code boundaries, so cross-correlation need not be zero. Multipath creates delayed copies that also disturb orthogonality. Unequal powers make residual interference from imperfectly orthogonal signals more damaging. Power control is therefore essential in practical CDMA uplinks: the base station aims to receive users at comparable power, preventing a nearby strong handset from masking a distant weak handset.

A conservative idealised user estimate is L simultaneous orthogonal users per code set, provided timing, code assignment and receiver assumptions hold. This is not a guaranteed capacity: real capacity is usually interference-limited and depends on target BER, voice activity, propagation and power control. For an engineering target BER $\leq 10^{-3}$, the code-length/user count must be swept under the intended channel model; this simplified experiment should not be interpreted as a universal user-capacity claim.

3.6 Conclusion

Longer spreading codes provide greater processing gain and more orthogonal code choices, while random codes introduce multiuser interference. Ideal synchronous Walsh orthogonality can suppress near-far interference mathematically, but practical timing errors and multipath make power control and synchronisation necessary.

Q4 – OFDMA: Giving Each User the Subcarriers They Can Use Best

4.1 OFDM fundamentals and orthogonality

For an OFDM system with subcarrier spacing Δf, useful symbol duration Tu = 1/Δf. For example, Δf = 15 kHz gives Tu = 66.67 µs. A cyclic prefix (CP) copies the end of the useful symbol to its beginning. If the CP is at least as long as the effective channel delay spread, it reduces inter-symbol interference and preserves subcarrier orthogonality under the assumed channel model. The CP consumes time and reduces net throughput.

For N subcarriers over useful duration Tu, the complex baseband subcarrier functions are exp(j2πkΔf t). Their inner product over Tu is zero for distinct integer subcarrier indices when ΔfTu = 1.

Subcarrier pair	Normalised inner product magnitude	Expected
k=0 and k=1	0 (analytical ideal)	Orthogonal
k=0 and k=2	0 (analytical ideal)	Orthogonal
k=3 and k=7	0 (analytical ideal)	Orthogonal

Numerical verification can be performed by sampling an integer number of useful symbol periods and computing the normalised complex inner product; finite precision should produce a value close to zero.

4.2 Allocation strategies and predictions

- Fixed contiguous blocks: each of eight users receives eight neighbouring subcarriers. It is simple and predictable, but a user may receive a poor part of the frequency response.
- Interleaved round-robin: each user receives every eighth subcarrier. It spreads frequency diversity more evenly and offers predictable fairness.
- Channel-aware best-subcarrier: each subcarrier is assigned to the user with the highest instantaneous rate on it. It should maximise sum throughput but can disadvantage users with consistently weak channels.

4.3 Simulation method

Eight users share 64 subcarriers. Each of 1,000 independent channel realisations uses independent unit-mean exponential power gains (Rayleigh fading power). Per-subcarrier rate is log2(1 + 10g) bits/s/Hz, corresponding to a fixed linear SNR factor of 10 and unit bandwidth normalisation. Fixed allocation gives eight contiguous subcarriers per user; interleaved allocation gives every eighth subcarrier; channel-aware allocation assigns each subcarrier to the user with the highest instantaneous rate. Jain's fairness index is (sum Ri)^2 / (U × sum Ri^2).

4.4 Results

Fixed contiguous: sum throughput = 186.789 normalised bit/s/Hz per realisation; Jain fairness = 1.0000.

User	1	2	3	4	5	6	7	8
Average throughput	23.341	23.368	23.420	23.401	23.258	23.540	23.326	23.136

Interleaved round-robin: sum throughput = 186.606 normalised bit/s/Hz per realisation; Jain fairness = 1.0000.

User	1	2	3	4	5	6	7	8
Average throughput	23.307	23.500	23.553	23.105	23.462	23.132	23.371	23.174

Channel-aware best-subcarrier: sum throughput = 299.984 normalised bit/s/Hz per realisation; Jain fairness = 0.9999.

User	1	2	3	4	5	6	7	8
Average throughput	37.798	37.532	37.572	37.186	38.220	37.522	37.149	37.004

4.5 Design decision and conclusion

A lecture hall should not maximise sum throughput without a minimum-rate constraint. A practical design is proportional-fair or weighted scheduling with a minimum-rate floor: first ensure each active user receives its minimum service, then allocate remaining subcarriers to users with favourable instantaneous channels. For a report-level design target, define a minimum average rate per user and verify it over many channel realisations; the exact target must be selected from the lecture's application requirements. The channel-aware maximum-rate policy can starve weak users because it repeatedly assigns resources to users with better channel gains. As the number of users increases, multiuser diversity can increase the best achievable sum rate, but fairness pressure and scheduling complexity also increase.

Q5 – SDMA: Reusing the Same Channel in Different Directions

5.1 ULA array factor derivation

Consider an N-element uniform linear array with element spacing d and progressive phase shift β. For a plane wave arriving at angle θ, the phase difference between adjacent elements is $\psi = kd\sin\theta + \beta$, where $k = 2\pi/\lambda$. The array factor is $AF(\theta) = \Sigma[n=0 \text{ to } N-1] \exp(jn\psi)$. For steering toward θ_o, choose $\beta = -kd\sin\theta_o$. The normalised magnitude is $|AF(\theta)|/N$.

For half-wavelength spacing $d = \lambda/2$, increasing N narrows the main lobe and increases directivity. For a uniformly excited ideal array, the first sidelobe level is approximately −13.3 dB for sufficiently large N; the exact sampled pattern and finite-array sidelobes are evaluated numerically.

5.2 SIR threshold and simulation

Define a 10 dB SIR threshold. With the beam steered to user A at broadside, the idealised SIR at user B is approximated as desired main-beam power divided by the array response power in B's direction. The numerical model uses a half-wavelength-spaced ULA, broadside steering and angles from 5° to 60°. This is a simplified single-path array-factor model; it excludes thermal noise, multipath, mutual coupling, calibration error and traffic interference.

N elements	Approximate HPBW (degrees)	Minimum tested separation meeting 10 dB
2	60.02	60°
4	26.36	30°
8	12.84	15°
16	6.36	10°

5.3 SIR results

N	Separation	HPBW (°)	Peak sidelobe (dB)	Idealised SIR at B (dB)
2	5	60.02	-150.00	0.08
2	10	60.02	-150.00	0.33
2	15	60.02	-150.00	0.74
2	20	60.02	-150.00	1.32
2	30	60.02	-150.00	3.01
2	40	60.02	-150.00	5.48
2	50	60.02	-150.00	8.89
2	60	60.02	-150.00	13.60
4	5	26.36	-11.30	0.41
4	10	26.36	-11.30	1.69
4	15	26.36	-11.30	4.00
4	20	26.36	-11.30	7.76
4	30	26.36	-11.30	150.00
4	40	26.36	-11.30	12.74
4	50	26.36	-11.30	11.49
4	60	26.36	-11.30	14.39
8	5	12.84	-12.80	1.78
8	10	12.84	-12.80	8.41
8	15	12.84	-12.80	29.13
8	20	12.84	-12.80	13.01
8	30	12.84	-12.80	150.00
8	40	12.84	-12.80	16.83
8	50	12.84	-12.80	31.43
8	60	12.84	-12.80	17.92
16	5	6.36	-13.15	8.57
16	10	6.36	-13.15	13.23
16	15	6.36	-13.15	29.18
16	20	6.36	-13.15	20.91
16	30	6.36	-13.15	150.00
16	40	6.36	-13.15	29.92
16	50	6.36	-13.15	31.61
16	60	6.36	-13.15	36.90

5.4 Inquiry and design decision

SDMA is limited by angular spread, users clustered in similar directions, multipath, finite antenna elements, side lobes, calibration errors and movement. Two users in nearly the same direction produce highly overlapping spatial signatures, so a single array cannot reliably separate them using direction alone. A toll plaza often has vehicles approaching along nearly the same direction; therefore SDMA should not be the only access method. Use sectorisation or spatial processing where lanes create angular separation, combined with time/frequency scheduling and handover/resource coordination.

5.5 Conclusion

The array-factor model shows that increasing the number of elements narrows the main beam and can improve spatial separation. However, users with nearly identical angles remain difficult to distinguish, and ideal pattern calculations overestimate real-world performance.

Q6 – Choosing Multiple-Access Techniques for Real Deployments

6.1 Weighted criteria

Five criteria are used. Ratings range from 1 (poor) to 5 (excellent). Weights sum to 100%. Coverage/capacity is weighted 25%, mobility support 20%, interference/jam resilience 20%, burst efficiency 20% and implementation simplicity 15%. Capacity and coverage receive the largest weight because all four deployments require reliable service at scale. Mobility, resilience and burst efficiency are also important; simplicity receives a lower but meaningful weight.

Criterion	Weight
Coverage/capacity	25%
Mobility support	20%
Interference/jam resilience	20%
Burst efficiency	20%
Implementation simplicity	15%

6.2 Weighted decision method

For each deployment, each access technique receives a rating from 1 to 5 for each criterion. Weighted score = Σ(weight × rating)/100. The ratings are engineering judgements for the stated use cases, not universal measurements; they should be interpreted alongside the evidence and limitations in Q2–Q5.

Wide-area cellular voice

Technique	Coverage/capacity	Mobility	Resilience	Burst efficiency	Simplicity	Weighted score
SDMA	4	4	3	3	2	3.30
TDMA	3	4	3	2	4	3.15
FDMA	2	3	3	2	5	2.85
CDMA	4	4	4	4	2	3.70
OFDMA	5	4	3	4	2	3.75

Recommended primary technique from this matrix: OFDMA.

Tactical jam-resistant link

Technique	Coverage/capacity	Mobility	Resilience	Burst efficiency	Simplicity	Weighted score
SDMA	2	3	2	2	2	2.20
TDMA	2	3	2	2	4	2.50
FDMA	2	2	2	2	5	2.45
CDMA	4	4	5	4	2	3.90
OFDMA	4	3	2	4	2	3.10

Recommended primary technique from this matrix: CDMA.

High-rate stadium downlink

Technique	Coverage/capacity	Mobility	Resilience	Burst efficiency	Simplicity	Weighted score
SDMA	4	3	3	4	2	3.30
TDMA	2	3	3	2	4	2.70
FDMA	2	2	3	2	5	2.65
CDMA	3	3	2	3	2	2.65
OFDMA	5	4	3	5	2	3.95

Recommended primary technique from this matrix: OFDMA.

5000 low-rate sensors

Technique	Coverage/capacity	Mobility	Resilience	Burst efficiency	Simplicity	Weighted score
SDMA	3	2	3	3	2	2.65
TDMA	4	3	3	4	4	3.60
FDMA	2	2	4	2	5	2.85

CDMA	4	3	4	4	2	3.50
OFDMA	3	3	3	3	2	2.85

Recommended primary technique from this matrix: TDMA.

6.3 Deployment recommendations

- Wide-area cellular voice: OFDMA as a flexible modern radio-access foundation, often combined with spatial reuse and scheduling. Legacy or specialised systems may use other combinations.
- Jam-resistant tactical link: CDMA-like spreading/frequency agility can provide processing gain and interference tolerance, but anti-jam performance depends on threat model, synchronisation, power control and cryptographic/network-layer protection.
- High-rate stadium downlink: OFDMA with SDMA/MIMO scheduling is appropriate because it can allocate frequency resources flexibly and exploit spatial separation.
- 5,000 low-rate sensors: TDMA or scheduled shared access with aggregation and sleep scheduling is attractive; dedicated FDMA per sensor is wasteful. A practical low-power network may use a hybrid random-access and scheduled mechanism.

6.4 Sensitivity check

Deployment	Base winner	Did any single weight ±20% change the winner?
Wide-area cellular voice	OFDMA	Yes
Tactical jam-resistant link	CDMA	No
High-rate stadium downlink	OFDMA	No
5000 low-rate sensors	TDMA	No

The sensitivity test changes one criterion weight at a time by −20% and +20%, then renormalises the weights. A winner change indicates that the decision is sensitive to subjective weighting and should be treated as a close decision. If no winner changes, the recommendation is more robust under this limited test. Changing all weights by the same percentage would not affect rankings because normalisation cancels the common factor.

6.5 Why real systems combine techniques

Real networks combine methods because each solves a different resource dimension. FDMA divides spectrum into frequency bands; TDMA divides time within a carrier; SDMA reuses resources spatially; CDMA separates users by spreading codes; OFDMA assigns orthogonal subcarriers. A hybrid can improve capacity and flexibility, but adds synchronisation, scheduling, signalling and implementation complexity. The least robust choice is the one whose performance depends most strongly on an assumption that the deployment cannot guarantee—for example, pure SDMA in a tightly clustered toll-plaza geometry or CDMA without effective power control.

Q7 – Diagnose, Redesign and Validate

7A. Diagnose the bottleneck

Selected limitation: dedicated resource reservation for low-duty-cycle telemetry, identified in Q2. A sensor report is active for only a small fraction of each reporting interval. If a dedicated channel remains reserved while the device is silent, most reserved resource-time is idle. The counted quantity that grows with the number of sensors is the total reserved resource-time, even though the actual transmitted payload remains small.

Illustrative scale: 5,000 devices each report once per minute. If each dedicated allocation is held for the full 60-second interval, the reservation is 5,000 × 60 = 300,000 resource-seconds per minute. If the radio is active for 6 seconds (10% duty cycle), actual active time is 5,000 × 6 = 30,000 resource-seconds. Under this idealised comparison, 90% of reservation time is avoidable. This calculation assumes each device has an individually reserved resource and ignores contention, guard time and aggregation.

7B. Design objective

Objective: reduce reserved resource-time for low-duty-cycle reports by at least 80% while preserving delivery reliability and meeting the application's reporting deadline. The redesign must avoid holding a dedicated resource during idle periods. Reliability, collision rate, queue delay and control overhead must be measured in a fuller implementation.

7C. Revised strategy

Use event-driven shared access with local buffering and scheduled aggregation. A sensor stores its report, wakes at an assigned or contention-controlled opportunity, requests/uses a short transmission opportunity, sends the report, and returns to sleep. A gateway can aggregate several reports where application semantics permit. Randomised backoff or assigned mini-slots reduce repeated collisions. The strategy exploits the known low duty cycle and the fact that payloads are short.

```
for each reporting interval:
    sensor_measurement = read_sensor()
    store_locally(sensor_measurement)

    wait_until_transmission_opportunity()
    if shared_channel_available():
        transmit_buffered_report()
        wait_for_acknowledgement()
        if acknowledgement_received():
            mark_report_delivered()
        else:
            retain_report_and_schedule_retry()
    else:
        apply_random_backoff()
        retain_report_for_next_opportunity()

gateway:
    receive_reports()
    validate_sequence_numbers()
    remove_duplicates()
    aggregate_if_allowed()
    forward_to_network()
```

New costs and trade-offs: the redesign needs buffering, scheduling or contention control, acknowledgements, retries and duplicate detection. It can introduce collision and queueing delay. It is best suited to delay-tolerant small reports; dedicated channels may remain preferable for continuous voice, strict low-latency control or guaranteed isolated links.

7D. Prediction before implementation

Prediction: under a 10% activity assumption, event-driven sharing should reduce resource reservation time by approximately 90% compared with holding one dedicated resource for the entire minute. This is a resource-time prediction, not a claim of 90% higher end-to-end throughput. Control overhead and collisions will reduce the realised gain.

7E. Prototype validation

Measure	Original dedicated reservation	Redesigned event-driven reservation
Devices	5000	5000

Reporting interval	60 s	60 s
Active transmit duration per report	6 s	6 s
Reserved resource-time per device	60 resource-s	6 resource-s
Total resource-time per 5,000 devices	300,000.0 resource-s	30,000.0 resource-s
Avoidable idle reservation	90%	0% in ideal active-only accounting

Calculated reduction in reserved resource-time = 1 − (30,000 / 300,000) = 90.0%. The redesigned column is an ideal lower-bound accounting of active resource use; a real prototype must add request, guard, acknowledgement, retry and collision overhead before claiming an achieved network gain.

7F. Final defence

The prediction is supported by the resource-time accounting: a 10% duty cycle means that continuous reservation wastes 90% of the reserved interval under the stated assumptions. The redesign introduces coordination and retry costs, so its practical benefit depends on report deadlines, contention load and control overhead. The original dedicated design can still be preferable for continuous or safety-critical traffic that needs predictable access. The engineering conclusion is not that shared access is always superior, but that resource reservation should match the traffic's actual activity pattern.

Overall Conclusion

The investigation shows that mobile and wireless describe different properties, and that mobile computing requires application support beyond radio connectivity. FDMA and TDMA can have equal nominal capacity but different utilisation under different traffic patterns. CDMA relies on code correlation, timing and power control. OFDMA can improve sum throughput through channel-aware allocation but must include fairness constraints. SDMA increases spatial reuse but is limited by angular separation and real antenna imperfections. A weighted decision process helps select access techniques according to deployment requirements rather than assuming one method is best for every network.

The Q7 redesign demonstrates how an engineering improvement should be derived from a measured bottleneck. Event-driven shared access can reduce idle reservation for intermittent telemetry, but adds scheduling, buffering and retry complexity. The final design must be validated against both efficiency and service requirements.

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Appendix A – Reproducible Python Code

The companion Python file contains the complete executable experiment code and can be run with Python 3 after installing NumPy and Matplotlib. It includes deterministic seeds and the models used to generate the report tables. Q6 uses an explicit weighted matrix; Q7 is a resource-time prototype calculation.

Install dependencies: pip install numpy matplotlib

Run: python simulation_code_Q1-Q7.py

Note: The report intentionally distinguishes simplified model outputs from real-world network guarantees. Replace the illustrative overhead assumptions and channel parameters with instructor-provided values if available.

Question	Assessed Area	Marks
Q1	Mobile communication vs. mobile computing	5
Q2	FDMA and TDMA	6
Q3	CDMA	7
Q4	OFDMA	6
Q5	SDMA	6
Q6	Comparison of SDMA, TDMA, FDMA, CDMA, OFDMA	7
Q7	Day-4 Design Challenge – Diagnose, Redesign & Validate	12
TOTAL		49